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The Honorable Mehmet Oz, MD
Administrator
Center for Medicare and Medicaid Services
U.S. Department of Health and Human Services
7500 Security Boulevard
Baltimore, MD 21244

RE: Request for Information on Health Technology Ecosystem (CMS-0042-NC)

The American Academy of Ophthalmology (the Academy) appreciates the opportunity to provide comments on the Request for Information on the Health Technology Ecosystem.

The Academy is the largest association of eye physicians and surgeons in the United States. A nationwide community of over 20,000 medical doctors, we protect sight and empower lives by setting the standards for ophthalmic education, supporting research, and advocating for our patients and the public. We innovate to advance our profession and to ensure the delivery of the highest-quality eye care.

We welcome the opportunity to highlight ophthalmology's unique role and needs within the evolving digital health landscape. Our specialty is a leader in leveraging Qualified Clinical Data Registries (QCDRs) to drive quality improvement, reduce reporting burden, and deliver timely, actionable data to physicians. However, their full potential remains underutilized within the broader health technology ecosystem, where they could play a greater role in advancing data-driven, patient-centered care.

Additionally, ophthalmology faces ongoing challenges in advancing digital workflows and achieving interoperability for electronic imaging. Despite the central role of imaging in our field, barriers to standardized image exchange across multiple companies limit the full potential of digital integration and thereby reduce efficiencies and savings within the broader health technology ecosystem.

We appreciate your consideration of these perspectives and look forward to continued collaboration to enhance the health technology ecosystem in ways that improve care quality, efficiency, and outcomes for patients nationwide.

TD-12: Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

QCDRs are essential tools for driving meaningful quality improvement, enabling digital quality measurement, and reducing administrative burden across the healthcare system. These registries enable physicians to participate in the Quality Payment Program (QPP) in ways that both reduce administrative burden and ensure reporting is meaningful to their clinical practice. The Academy and other medical specialty societies have long championed improvements in patient care by operating QCDRs, efforts that are typically financed by the societies themselves.

Despite their tremendous potential to advance quality measurement, improve outcomes and support the shift toward value-based care, QCDRs remain significantly underutilized.

As you are aware, the Medicare Access and CHIP Reauthorization Act of 2015 (MACRA; P.L. 114-10) requires the Secretary of Health and Human Services to encourage the use of QCDRs and certified electronic health record technology for reporting measures under the quality performance category of the Merit-Based Incentive Payment System (MIPS) program.[1]

QCDRs provide timely and actionable feedback to physicians by developing performance benchmarks and sharing these benchmarks with registry participants.[2] The study and use of registry data enables providers to identify best clinical practices in real-time and detect deficiencies or disparities in care that require corrective action.

For example, The American Academy of Ophthalmology IRIS® (Intelligent Research In Sight) Registry has fully developed 30 QCDR quality measures, 14 of which are currently available for reporting in MIPS. *All* these measures are classified by the Centers for Medicare & Medicaid Services (CMS) as outcome or intermediate outcome measures as they are directly linked to improvements in patient outcomes specific to eye care. These include measures focused on cataract surgery outcomes and the management of chronic eye diseases such as glaucoma, diabetic eye disease, and age-related macular degeneration.

For highly specialized fields such as ophthalmology with several subspecialties, specialty- and sub-specialty specific QCDR measures are the most effective tools for identifying and acting on quality improvement opportunities. These measures capture the nuances of care across ophthalmic subspecialties that can meaningfully support continuous improvement in patient outcomes. Measures that focus on more broad, preventive care, while important to improving quality, offer limited relevance or utility to ophthalmologists and their patients.

Clinical data registries also serve as invaluable sources of real-world data (RWD) and are key partners in generating real-world evidence (RWE) for best health care. For example, the IRIS Registry has amassed data from more than 788.48 million patient visits and 84.87 million unique patients, collected from approximately 14,925 clinicians across 2,967 practices. The collection of this data enables us to identify and measure meaningful quality improvement opportunities that would otherwise go unmeasured and unimproved.

Registry data presents a remarkable opportunity for knowledge discovery and collaborative research. It can inform the natural history of disease, track the prevalence of rare conditions, monitor technology adoption, analyze comparative effectiveness, and more—all in real-world settings. To date, over 160 peer-reviewed publications have been published using IRIS Registry data. Imagine the possible benefits with future registry integration across specialties if QCDRs continue to be supported.

Despite these clear benefits, QCDRs continue to be underrecognized and underutilized by CMS in MIPS and other federal quality reporting programs. As currently implemented by CMS, QCDRs are treated merely as an optional reporting mechanism.

Given its statutory directive to encourage the use of QCDRs, CMS must do more to incentivize the QCDR-reporting option to realize their full potential within the QPP framework. As CMS considers ways to modernize and digitize quality reporting, it must recognize that QCDRs are uniquely positioned to bridge clinical care with quality measurement if it truly intends to achieve the broader goal of improving healthcare quality and patient outcomes.

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[1] MACRA, Pub. L. No. 114-10, § 101(c), 129 Stat. 87 (2015)

[2] Rich WL 3rd, Chiang MF, Lum F et al. Performance rates measured in the American Academy of Ophthalmology IRIS Registry (Intelligent Research in Sight). Ophthalmology. 2018;125:782-784.

PR-8: What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

One way that CMS can simplify quality reporting requirements and incentivize QCDRs is by awarding multi-performance category in MIPS. Currently, there is conceptual overlap among the different MIPS performance categories: quality, cost, improvement activities, and promoting interoperability.

For example, practices that integrate their electronic health record (EHR) systems with a QCDR like the IRIS Registry should receive credit across all applicable MIPS categories. Such an approach would not only acknowledge the quality efforts made by physicians, but also significantly reduce the administrative burden of duplicative reporting requirements.

VB-3: What are essential health IT capabilities for value-based care arrangements? What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?

Most existing value-based care arrangements do not allow for ophthalmologists' and other specialists' participation in any significant portion of their care delivery. To address this gap, CMS should dedicate resources to developing and implementing value-based care models that are inclusive of clinicians practicing outside hospital settings and that meaningfully assess specialty care.

To achieve this, we urge the administration to leverage QCDRs and their specialty-specific measures as a foundation for these efforts. As discussed previously, QCDRs are uniquely positioned to evaluate the quality of specialty care through clinically relevant, outcomes-focused measures. They also offer the necessary health information technology (IT) infrastructure to support data collection, real-time performance feedback, and continuous quality improvement.

QCDRs also provide a meaningful and scalable pathway for specialty physicians to engage in value-based care arrangements—one that reflects the complexity and nuance of their clinical practice. **We strongly encourage CMS to collaborate with medical specialty societies in the design and implementation of these models to ensure they are tailored to the needs of specialists and aligned with high-quality patient care.**

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem?

As with all clinical data, imaging information must align with national interoperability priorities to reduce inefficiencies and improve patient outcomes. This is especially important in ophthalmology, where imaging plays a central role in diagnosis, disease monitoring, and treatment.

To support fully integrated digital workflows, the Academy emphasizes the importance of compliance with DICOM (Digital Imaging and Communications in Medicine)—a widely recognized standard for imaging data. DICOM compliance enables the exchange of structured information such as patient demographics, device specifications, and discrete imaging measurements, making it the gold standard for ocular imaging devices.[3]

While progress has been made, there remains a significant opportunity to improve universal DICOM adoption among ophthalmic imaging manufacturers. This underutilization continues to hinder the realization of fully integrated digital workflows in ophthalmology.

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[3] Lee AY, Sadda S, Lum F. Joint American Academy of Ophthalmology and Association for Research in Vision and Ophthalmology Policy Statement: The Time for Digital Imaging Standards Implementation Is Now. Ophthalmology. 2022 Nov;129(11):1229-1231. doi: 10.1016/j.ophtha.2022.07.034. Epub 2022 Sep 10. PMID: 36096893.

Manufacturers remain hesitant to participate in the standards development process or to implement existing standards in their products due to concerns over the increased costs associated with DICOM compliance, the limited understanding among physician purchasers regarding the benefits of DICOM, and the absence of regulatory mandates requiring adherence to these standards. Additionally, the lack of third-party certification for DICOM conformance statements raises concerns about the extent of comprehensive compliance across the industry.[4]

Federal agencies have taken meaningful steps to address these challenges. Notably, the National Eye Institute, Food and Drug Administration, and Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology convened a 2022 workshop to engage manufacturers and explore strategies for advancing DICOM adoption in ophthalmology. While promising, further action is needed to develop incentives, remove implementation barriers, and provide the education and regulatory clarity necessary to achieve widespread compliance.

The Academy encourages CMS and other agencies to take steps within their authority to accelerate imaging interoperability in ophthalmology, ensuring that providers can access high-quality, timely, and comprehensive imaging data essential to delivering patient-centered care and supporting the broader goals of the digital health ecosystem. The Academy remains committed to partnering with stakeholders to achieve this vision.

To advance DICOM adoption, the Academy recommends federal agencies to consider policy and programmatic approaches, such as requiring DICOM-conformant devices in federal purchasing and research, incorporating DICOM standards into FDA approvals, and incentivizing physician adoption to promote consistent, high-quality care.

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The Academy stands ready to work with CMS to advance a more connected, data-driven healthcare system that supports high-quality, medically necessary eye care for all patients. We welcome the opportunity to collaborate on solutions that strengthen digital infrastructure, expand interoperability, and realize the full potential of tools like QCDRs.

If you have any questions or to schedule a meeting to discuss our shared concerns further, please contact the Academy's Director of Health Policy, Brandy Keys, MPH, at bkeys@aao.org or via phone at 202-737-6662.

Sincerely,

Michael X. Repka, MD, MBA
President
American Academy of Ophthalmology

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[4] Lee, A. Y., Sadda, S., Lum, F., et al. (2022). Joint American Academy of Ophthalmology and Association for Research in Vision and Ophthalmology policy statement: The time for digital imaging standards implementation is now. Ophthalmology, 129(11), 1229-1231. https://doi.org/10.1016/j.ophtha.2022.07.034